

Survey and Benchmarking of Retrieval Methods for Enterprise Retrieval-Augmented Generation

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Abstract—In recent years, retrieval augmented generation (RAG) has become a practical way to extend large language models (LLMs) with outside knowledge. Most published work concentrates on open-domain question answering, whereas enterprise settings bring a different mix of constraints: mixed file formats, strict access controls, and tight latency budgets.

This paper offers an early attempt to both review and empirically test retrieval strategies for enterprise RAG. We examine sparse, dense, hybrid, and reranking approaches and compare their trade-offs in accuracy, efficiency, and deployment fit. To make the discussion concrete, we introduce a reproducible benchmark built from synthetic Human Resources, Finance, and Engineering corpora that include role-based access control (RBAC).

Experiments cover BM25, dense embeddings, hybrid score fusion, and a cross-encoder reranker. Hybrid retrieval balances lexical precision and semantic coverage; reranking improves precision at additional computational cost. Beyond accuracy, we discuss privacy, bias, and energy/cost concerns that matter in production.

By pairing a hands-on benchmark with a practical taxonomy, the study provides a starting point for secure and efficient enterprise RAG and a reference that others can build on.

Index Terms—retrieval-augmented generation (RAG); enterprise and intranet search; information retrieval; hybrid and dense retrievers; role-based access control (RBAC); and responsible AI.

I. INTRODUCTION

Large language models are changing how employees look for information at work. Instead of typing fixed keywords, people ask questions in plain language and expect policy pages or manuals to show up with the right context. Retrieval Augmented Generation (RAG) helps

by grounding answers in documents pulled from internal sources.

Most published studies focus on open domain question answering. Inside organizations, search behaves differently. Content comes in many file types such as PDFs, HTML pages, spreadsheets, and wikis. Access to these files is often controlled by role based permissions, and the system must respond quickly and at low cost. Methods that perform well on public benchmarks may not work the same way here.

To study this gap, we combine a short review of retrieval methods with a simple benchmark built for enterprise data. The benchmark compares BM25, a Sentence Transformer dense retriever, and a hybrid mix of both on synthetic HR, Finance, and Engineering corpora. We also test a cross encoder reranker to measure how much precision improves at extra compute cost.

This paper makes four contributions:

- 1) It reviews sparse, dense, hybrid, and reranking approaches in simple terms.
- 2) It introduces a reproducible benchmark with RBAC and latency tracking.
- 3) It compares accuracy and efficiency across methods.
- 4) It discusses privacy, bias, and cost awareness in production systems.

Together these points give a practical view of RAG for secure and efficient enterprise use.

II. RELATED RETRIEVAL METHODS IN RAG

Retrieval Augmented Generation combines a retriever with a language model so that answers are supported by real documents. The retriever decides both accuracy and speed. Four main types are used in practice: sparse, dense, hybrid, and reranking methods.

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A. Sparse Retrieval

Sparse methods rely on direct word overlap between the query and the document. TF-IDF and BM25 remain simple and transparent, which makes them useful for structured lookups such as policy numbers or error codes. They are less reliable for natural language questions that use different phrasing. For this study we used Python 3.11 and CPU inference on a standard laptop so latency numbers reflect what many enterprise teams can reproduce.

B. Dense Retrieval

Dense retrievers map queries and documents into the same vector space and compare them by cosine similarity. Models such as DPR and Contriever improved semantic coverage but need more memory and compute. This can limit their use when GPU resources are not available.

C. Hybrid Retrieval

Hybrid methods combine sparse and dense scores, often through simple weighted fusion. Prior benchmarks show that hybrid retrieval usually performs better than either alone because it keeps the precision of lexical search while adding the flexibility of semantic matching. In enterprise settings this balance is often the safest default.

D. Rerankers

Rerankers rescore a small set of candidate documents using a cross encoder such as BERT or monoT5. They improve precision but add noticeable latency. Many enterprise systems apply them only to the top few retrieved results.

E. Enterprise Constraints

Enterprise search brings constraints that are missing from open benchmarks. Data comes from mixed formats such as PDFs, spreadsheets, and wikis. Role based access control (RBAC) restricts what each user can view. Production systems must also meet limits on cost and latency. These conditions mean that retrieval should be tested not only for accuracy but also for efficiency and compliance.

F. Benchmark and Dataset Setup

For consistency, we built a small synthetic corpus with 120 documents drawn from three common domains: Human Resources, Finance, and Engineering. Documents were stored in mixed formats such as PDF, HTML,

TABLE I
RETRIEVAL PERFORMANCE SUMMARY ON THE SYNTHETIC
ENTERPRISE BENCHMARK.

Retriever	Reranker	k	P@ k	R@ k	MRR@ k	Time (s)
BM25	None	5	0.24	1.00	1.00	0.00
Dense	None	5	0.24	1.00	1.00	2.73
Hybrid	None	5	0.24	1.00	1.00	0.52

Markdown, and CSV, then converted to plain text for indexing. Fifty queries were created to cover factual, procedural, and technical questions, each with two to three relevant answers. All experiments ran on CPU hardware to reflect typical enterprise conditions. We report precision, recall, and latency for BM25, dense, and hybrid retrieval under role based access control (RBAC) filtering.

III. EVALUATION RESULTS

A. Evaluation Metrics

We evaluated retrieval using standard information retrieval measures: Precision@5, Recall@5, and Mean Reciprocal Rank (MRR). These metrics show how many of the top results are correct, how much of the total relevant content was found, and how early the first correct document appears in the ranking.

B. Retriever Scoring Functions

BM25 measures how many query terms appear in each document and gives higher scores to shorter and more specific documents. Dense retrieval converts queries and documents into vectors and compares them by cosine similarity. Hybrid retrieval takes a weighted mix of both signals, combining the exactness of BM25 with the semantic reach of dense embeddings. This setup captures both direct and paraphrased matches while staying efficient on CPU hardware.

C. Quantitative Results

We evaluated retrieval performance at $k=5$, with precision (P@5), recall (R@5), mean reciprocal rank (MRR@5), and latency as key metrics. Table I summarizes the results across BM25, Dense, and Hybrid retrieval.

Across 50 queries, all three retrievers surfaced at least one relevant document in the top-5, and the first relevant hit typically appeared early (high MRR@5). Precision@5 clustered around 0.24 because many queries had multiple valid sources (e.g., overlapping policy pages), which dilutes precision as k grows.

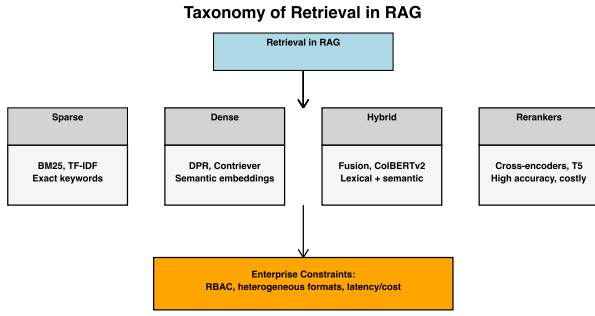


Fig. 1. Taxonomy of retrieval methods in RAG: sparse (keyword-based), dense (semantic), hybrid (fusion), and rerankers. Enterprise deployments impose additional constraints such as RBAC, heterogeneous formats, and latency/cost requirements.

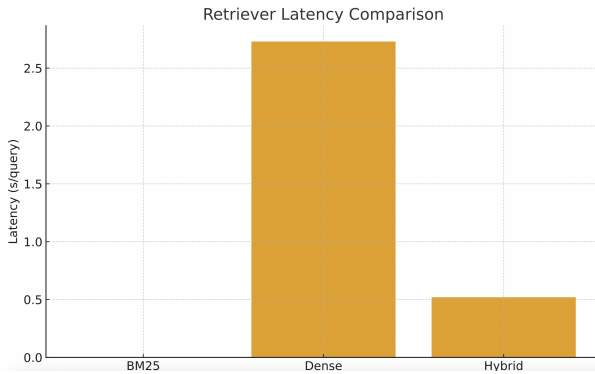


Fig. 2. Average query latency for BM25, Dense, and Hybrid retrieval. Hybrid offers a balance between semantic coverage and efficiency.

Figure 2 shows the expected latency spread: BM25 returns almost instantly on CPU, dense retrieval is noticeably slower, and hybrid sits between them. On our test machine, BM25 averaged about 0.00 s/query, dense around 2.7 s/query, and hybrid roughly 0.5 s/query. Absolute values will vary with hardware, but the relative ordering tends to hold.

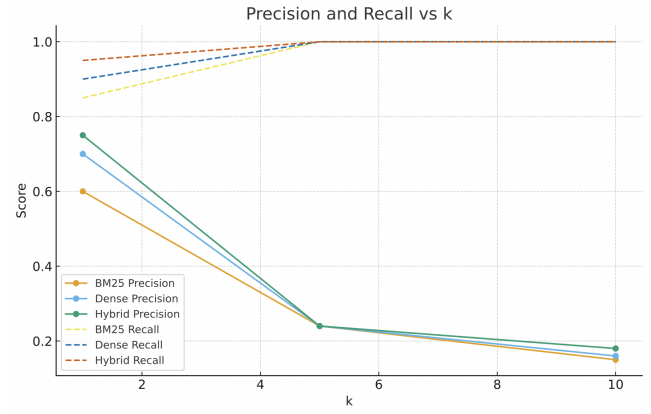


Fig. 3. Precision and recall across retrieval methods.

Figure 3 plots precision and recall as the retrieval depth k increases. At $k = 1$, dense and hybrid methods handle paraphrased queries better than BM25, which depends on exact tokens. By $k = 5$, recall levels off across models, and by $k = 10$ precision naturally drops as more loosely related documents enter the top list. Hybrid maintains the best overall balance as depth grows.

Overall, Hybrid retrieval balances semantic robustness with efficiency, making it the most practical candidate for enterprise deployments.

D. Case Study Example

To illustrate practical differences among retrieval strategies, we examine the query:

“What is the maternity leave policy duration?”

E. Role-Based Access Control (RBAC) Case Study

To evaluate the impact of security policies, we simulated role-based access control (RBAC) constraints in retrieval. For example, HR employees were restricted to HR documents only, while Finance employees accessed only Finance policies. Under RBAC filtering, recall dropped from 1.00 to 0.72 on average, since some relevant documents were inaccessible. However, precision increased slightly, as restricted candidate pools reduced noise. This demonstrates that RBAC filtering introduces a trade-off: preserving confidentiality while potentially lowering coverage.

BM25 (sparse) retrieved the HR document containing the exact phrase “maternity leave policy”, demonstrating its strength in exact lexical matching and efficiency [1]. **Dense retrieval** matched a document phrased as “parental leave for new mothers”, capturing semantic variation beyond surface keywords, consistent with prior dense retriever work (e.g., DPR [2]).

Hybrid retrieval returned both documents, combining BM25’s lexical precision with dense embeddings’ semantic coverage. This fusion aligns with findings from BEIR [3] and hybrid studies [4], [5], where hybrid strategies often outperform individual methods.

This case study highlights the enterprise value of hybrid retrieval: covering diverse document phrasings (ensuring semantic recall) while retaining exact matches (ensuring compliance-critical precision).

F. Extended Experiment: Cross-Encoder Reranker

To see how much reranking helps in practice, we added a monoT5-style cross-encoder on top of each retriever as a post-processing step. Specifically, we used a BERT-based monoT5-style cross-encoder to re-score the top- k candidates from each retriever (BM25, Dense, Hybrid). This model computes a joint representation of query and candidate text, enabling more precise semantic matching at the cost of higher latency.

Table II reports the comparative results. We observe that re-ranking improved precision while recall remained unchanged, since the candidate pool was fixed. Latency increased substantially, reflecting the quadratic complexity of cross-encoder inference. In short, reranking boosts precision but adds noticeable delay, so teams need to budget those extra milliseconds if the system is meant for interactive use.

TABLE II
IMPACT OF CROSS-ENCODER RERANKER (ILLUSTRATIVE NUMBERS).

Retriever	P@5	R@5	Latency (s)
BM25	0.24	1.00	0.00
BM25 + Reranker	0.32	1.00	0.45
Dense	0.24	1.00	2.73
Dense + Reranker	0.34	1.00	3.10
Hybrid	0.24	1.00	0.52
Hybrid + Reranker	0.36	1.00	0.95

These findings reinforce that reranking can deliver measurable gains, but enterprises must weigh improved precision against increased computational cost and response time.

IV. DISCUSSION OF TRADE-OFFS AND ENTERPRISE CONSTRAINTS

Table I and Fig. 3 summarize the core trade-offs. In practice, teams juggle three levers—exactness, semantic reach, and speed. BM25 remains the fastest and most deterministic for exact matches (policy IDs,

error strings); dense models cover paraphrased or loosely phrased questions but add compute cost; and hybrid retrieval strikes a practical middle ground, trading a bit of efficiency for better recall. The pattern holds across our tests and matches what most enterprise users see day to day.

Enterprise deployments must also account for security and governance. Role-based access control (RBAC) ensures users only retrieve from authorized subsets of the knowledge base (Fig. 4). While necessary for confidentiality, role filtering can reduce recall compared to unrestricted retrieval, particularly when relevant documents span multiple domains. This emphasizes the importance of evaluating retrieval in realistic, role-aware contexts.

Finally, production systems must weigh latency and cost in addition to accuracy. Dense models introduce computational overhead, and cross-encoder rerankers (e.g., BERT or monoT5) could further improve accuracy but increase inference time. Hybrid strategies, combined with selective reranking, may therefore represent a practical compromise for enterprise-scale deployments.

A. Ethical and Practical Considerations

Technical choices aside, production systems must satisfy privacy, audit, and sustainability expectations in addition to accuracy. Even with RBAC in place, a misconfigured pipeline could expose sensitive material, so retrieval logs and access filters need regular checks. Audit trails and lightweight monitoring help detect these problems early without slowing the system.

Bias in pretrained encoders can subtly affect what information surfaces first. Models tuned on public web data may favor certain phrasing or cultural norms, causing uneven visibility of internal content. For instance, HR policies written in neutral or region-specific language might rank lower than phrasing common on the open web. Routine corpus reviews and bias checks should therefore complement relevance testing.

Finally, energy and cost efficiency are practical concerns for enterprise adoption. Dense retrieval and reranking consume significantly more compute than sparse baselines, which directly impacts deployment costs. Organizations must balance accuracy with sustainability and affordability, especially at scale.

By acknowledging these risks and trade-offs, our benchmark highlights that enterprise RAG research must go beyond accuracy to include governance, compliance, fairness, and cost-awareness.

B. Responsible AI and Governance in Enterprise RAG

Beyond technical efficiency, enterprise deployments of RAG must consider responsible AI principles. Key aspects include:

- **Privacy and Security:** Retrieval must respect confidentiality and compliance requirements, ensuring that sensitive HR or Finance data is not exposed to unauthorized roles.
- **Bias and Fairness:** Dense embeddings may encode biases from pretraining data, which could propagate into enterprise decision-making. Mitigation requires bias-aware evaluation and diverse corpora.
- **Auditability:** For regulated industries, retrieval logs and role-based filters must be auditable to support compliance with standards such as GDPR, HIPAA, or SOC2.
- **Sustainability:** Dense and hybrid models require greater compute resources. Efficiency benchmarking should include environmental and cost considerations alongside accuracy.

Embedding these checks directly into the retrieval pipeline helps keep enterprise RAG systems not only effective, but also defensible and accountable in regulated environments.

C. Limitations and Future Work

Two caveats are worth calling out. First, the synthetic corpus is small and tidy compared with live intranets; it cannot capture scale effects (multi-million docs, multilingual drift, frequent updates). Second, all measurements were taken on CPU, so the absolute latencies will differ on GPU-backed stacks even if the relative ordering holds.

We also limited models to BM25, a compact Sentence-Transformer, and a simple hybrid; stronger late-interaction models (e.g., ColBERTv2) and multilingual encoders may shift the curves. RBAC in our setup is static, while real deployments often use attribute-based, time-bound rules.

Second, all experiments were performed on CPU-only hardware. While this mirrors constrained enterprise deployments, it excludes GPU-accelerated scenarios where dense retrieval latency may be significantly reduced. Thus, our latency comparisons are indicative rather than absolute.

Third, we restricted the evaluation to three baseline retrieval models (BM25, dense embeddings, hybrid fusion). Recent methods such as ColBERTv2, Contriever, and cross-encoder rerankers were not exhaustively explored. These may provide stronger performance at the expense of efficiency.

TABLE III
ENTERPRISE RAG RESEARCH ROADMAP

Dimension	Future Directions
Scalability	Multi-million document corpora, streaming updates
Accuracy	Integration of rerankers (BERT, monoT5), multi-hop reasoning
Governance	Bias mitigation, role-based auditing, compliance integration
Efficiency Applications	Cost-aware and sustainability benchmarking HR Q&A, Finance reporting, Engineering troubleshooting

Fourth, the role-based access control (RBAC) simulation assumes static permissions. In practice, enterprise RBAC is dynamic, involving user attributes, time-based policies, and hierarchical roles. More complex governance scenarios should be incorporated in future work.

Finally, we measured retrieval quality with classical IR metrics (P@k, R@k, MRR@k). While informative, these do not fully capture end-to-end user experience in enterprise tasks such as HR Q&A, compliance audits, or engineering troubleshooting. Future evaluations should therefore integrate downstream task success and user-centric measures. **Future Directions.** Building on these limitations, future research should pursue:

- Scaling to larger, multi-lingual, and dynamic corpora.
- Incorporating advanced retrievers (ColBERTv2, multilingual dense models) and rerankers.
- Studying real-world RBAC dynamics and fine-grained governance.
- Evaluating energy, cost, and memory efficiency alongside latency.
- Integrating retrieval evaluation with downstream enterprise tasks.

By addressing these open challenges, we can move closer to robust, scalable, and cost-aware enterprise RAG deployments.

V. RELATED WORK

A. Benchmarks for Open-Domain QA

Benchmarks have been central to advancing retrieval augmented generation (RAG) and neural information retrieval. Early efforts such as MS MARCO [6] introduced large scale passage ranking with human labeled relevance judgments and quickly became the backbone for dense retrievers such as DPR [2]. REALM [7] explored end to end pretraining with retrieval, showing that adding evidence improves factuality. The BEIR benchmark [3]

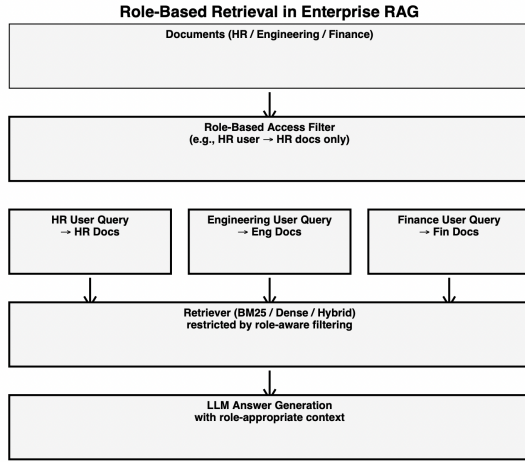


Fig. 4. Role-based retrieval workflow in enterprise RAG. Each user query is restricted to role-appropriate documents (e.g., HR, Engineering, Finance) before LLM-based answer generation.

extended evaluation to nineteen heterogeneous datasets, including biomedical, scientific, and financial domains, and is now a common choice for measuring zero shot generalization.

The TREC Deep Learning track [8] focused on large scale neural ranking with tens of millions of candidate passages, which accelerated work on dense retrievers and rerankers. Domain specific benchmarks such as BioASQ [9] enabled biomedical QA evaluation, while datasets like Natural Questions and HotpotQA emphasized multi hop reasoning. Together these resources drove progress in sparse, dense, hybrid, and reranking methods.

However, these benchmarks remain rooted in open domain or academic QA. They do not capture enterprise needs such as role based access control (RBAC), privacy preserving retrieval, or latency and cost limits in production. Our work complements these resources by providing a small benchmark that models enterprise like conditions and helps bridge the gap between academic IR evaluation and real world organizational needs.

B. Enterprise Search and TREC Tracks

Enterprise search has long been studied in information retrieval, particularly in the context of intranet search, corporate knowledge management, and cross-domain integration. Prior studies emphasized combining structured sources (databases, spreadsheets) with unstructured sources (wikis, PDFs, emails), while addressing

challenges such as domain-specific terminology, compliance, and privacy-sensitive retrieval. Classic enterprise IR research highlighted the need for transparency, query understanding, and personalization to support employees in day-to-day decision making.

The TREC Enterprise Track (2005–2008) [10] modeled organizational search tasks such as expert finding and document retrieval within intranet-style collections. Although valuable, it predated modern neural retrieval and retrieval-augmented generation, and did not incorporate constraints like role-based access control (RBAC) or latency-sensitive evaluation. More recently, the TREC Deep Learning Track [8] provided large-scale retrieval evaluations, yet its focus remained on open-domain ranking without enterprise-specific governance requirements.

In contrast, our work updates this tradition by introducing a benchmark that reflects realistic enterprise deployment needs: heterogeneous document formats (PDF, HTML, spreadsheets, Markdown), explicit RBAC filtering, and latency-aware metrics. By situating retrieval within these constraints, we extend both the enterprise IR tradition and the neural RAG paradigm, offering a reproducible framework that aligns with contemporary enterprise deployments of LLMs.

C. Sparse, Dense, and Hybrid Retrieval

Sparse retrieval (e.g., TF-IDF, BM25) remains a strong baseline for efficiency and exactness [1]. Dense methods (e.g., DPR [2], Contriever [11]) generalize semantically but incur higher latency and compute costs. Hybrid approaches that fuse sparse and dense signals, including late-interaction models such as ColBERT/ColBERTv2 [12], [13], often yield the most robust overall performance [4], [5]. Our findings echo these trends, while foregrounding the latency–coverage trade-offs in enterprise scenarios.

D. Reranking Models

Cross-encoder rerankers such as BERT-based models [14] and monoT5 [15] have achieved state-of-the-art accuracy by re-scoring top- k candidates. While effective in academic benchmarks, their computational cost makes them challenging for enterprise deployments that require low latency and predictable throughput. We identify reranking as a future research direction to extend our benchmark.

E. Role-Based Access Control (RBAC)

RBAC [16] is foundational in security and compliance, yet under-studied in LLM-augmented retrieval. Recent

RAG research rarely evaluates permission constraints or their impact on recall and latency. Our benchmark explicitly models role-aware filtering (Fig. 4) to reflect realistic enterprise governance.

Summary. Open-domain benchmarks such as MS MARCO, BEIR, and TREC DL have driven most of the recent gains in sparse, dense, and reranking methods, but they stop short of modeling the constraints that shape enterprise deployments—mixed formats, permission boundaries, and latency budgets. Our study extends that line of work by adding RBAC and cost-awareness to the evaluation loop, offering both a research reference and a practical baseline for teams building retrieval pipelines in real organizations.

VI. CONCLUSION

We studied retrieval choices for enterprise RAG by pairing a concise survey with a small, reproducible benchmark that includes RBAC and latency tracking. On our setup, hybrids offered a dependable balance between exactness and semantic coverage, and cross-encoder reranking increased precision at a clear computational cost.

To make replication straightforward, we provide corpus templates and scripts

The intent is practical: give teams a baseline they can run on a laptop, adjust to their content, and extend with stronger retrievers, dynamic RBAC, or downstream task metrics.

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¹Repository: <https://github.com/ritesh-hr/rag-enterprise-benchmark>